

Asymmetric Interdependence and the Limits of Multivectorism: Kazakhstan–China Economic Relations, 2015–2025

Abstract

This paper examines the structural evolution of Kazakhstan–China economic relations between 2015 and 2025, arguing that the bilateral relationship has transitioned from relative trade symmetry to accelerating asymmetric interdependence. Drawing on bilateral trade data, sectoral composition analysis, and three quantitative indices (the Trade Intensity Index, the Revealed Comparative Advantage index, and a bilateral Trade Asymmetry Index), the paper documents China’s emergence as Kazakhstan’s dominant import partner, a persistent commodity-manufacture divide in bilateral product flows, and the reversal of Kazakhstan’s modest bilateral trade surplus into deficit by 2023. Theoretically, the paper engages the tension between Keohane and Nye’s complex interdependence framework and Hirschman’s asymmetric dependence thesis, testing which interpretation the post-2022 empirical record supports. Kazakhstan has preserved formal multi-vector diplomatic autonomy, but the structural foundations of that autonomy are eroding under the combined pressure of Chinese import penetration, investment concentration, and the geopolitical displacement of Russia as a regional trade intermediary. Diplomatic mechanisms alone have proven insufficient to arrest these structural trends in the absence of productive diversification targeting non-commodity export sectors.

Keywords: Kazakhstan, China, Belt and Road Initiative, asymmetric interdependence, multivectorism, trade intensity, revealed comparative advantage, economic diplomacy

1 Introduction

Between 2015 and 2025, Kazakhstan’s bilateral trade with China nearly tripled in absolute value while the balance shifted from surplus to deficit. China’s share of Kazakhstan’s total import bill grew from under 20% to over 27%, and Chinese automotive, electronic, and industrial goods filled market space vacated by European and Russian suppliers following the 2022 invasion of Ukraine. This reconfiguration sits in direct tension with Kazakhstan’s foundational foreign policy doctrine of multivectorism, which holds that the republic should maintain diversified, balanced engagement with all major powers in order to preserve sovereignty and resist subordination to any single partner.

The central question this paper addresses is the following: to what extent has the deepening of Sino-Kazakh economic relations between 2015 and 2025 produced structural asymmetric dependence, and what does the empirical record reveal about the efficacy of Kazakhstan’s diplomatic and policy instruments in moderating that dependence? The question carries significance on two levels. It tests the boundary conditions of asymmetric interdependence theory in a case where the smaller state possesses geographic leverage and diverse diplomatic portfolios. It also generates policy-relevant findings for other small and medium states navigating economic integration with a dominant industrial partner under conditions of commodity export dependence, a situation shared broadly across Central Asia and along BRI corridors.

The paper is organized as follows. Section II reviews the relevant theoretical and empirical literature. Section III describes the data and analytical framework. Section IV analyzes Kazakhstan’s external trade structure and China’s rising share within it. Section V examines the structural features of bilateral trade, including sectoral composition, the commodity-manufacture divide, and the accelerating deficit. Section VI compares the China relationship to Kazakhstan’s partnerships with Russia and the European Union. Section VII synthesizes the findings and draws theoretical and policy conclusions.

2 Literature Review

2.1 Theoretical Framework

The theoretical architecture of this paper draws on two intersecting bodies of work. Geoeconomics, as articulated by Luttwak (1990) and operationalized by Blackwill and Harris (2016), holds that interstate competition increasingly operates through commercial instruments, with capital deployment and market access substituting for military coercion. Blackwill and Harris identify China as the leading contemporary practitioner of this approach, deploying infrastructure finance, preferential investment, and institutional frameworks in the service of strategic objectives. Applied to Central Asia, Rolland (2017) and Haji and Khattak (2025) characterize the BRI along these lines, as a grand strategy that creates structured economic dependencies redeemable as political leverage rather than a straightforward development program.

The second framework is complex interdependence and its asymmetric variant. Keohane and Nye (1977) argue that states embedded in multiple overlapping channels of exchange face prohibitive exit costs even when relationships are unequal, because dependence runs simultaneously across energy, finance, logistics, and consumer markets. Hirschman’s (1945)

asymmetric dependence thesis introduces the key qualification: when one state has substantially more alternative partners than the other, the structurally stronger party accumulates political leverage that extends beyond the economic domain. The empirical question this paper addresses is which characterization better fits Kazakhstan’s position in 2025, whether it retains sufficient diversification to sustain genuine autonomy or whether the terms of its engagement with China have narrowed its alternatives to the point where leverage has become structurally embedded.

2.2 The BRI and Kazakhstan

Kazakhstan holds a central position in BRI scholarship both because Xi Jinping chose Astana for the initiative’s 2013 announcement and because the country’s geography is indispensable to the overland Silk Road corridor. Lim, Tjia, and Murashkin (2025) advance three arguments in the most comprehensive post-BRI bilateral trade study to date: Kazakhstan has avoided structural dependency on the Chinese economy; trade has shifted from a natural-resources focus toward a more balanced mix including intermediate goods and consumer products; and the future trajectory anticipates continued growth with bottlenecks. This paper’s empirical findings engage the first two claims directly, since the 2023–2025 data complicate both. Castellano and Castillo (2024) find that China’s role in Kazakhstan’s downstream petrochemical development varies by project according to the specific combination of BRI drivers at play, suggesting that the agency-structure balance is sector-specific rather than uniform. The Middle Corridor has attracted growing scholarly attention (Pepe, 2021; GFSIS, 2025), and the February 2026 launch of a China–Europe freight service via the Trans-Caspian route confirms Kazakhstan’s deepening transit centrality, even as the disruption of the Russian corridor has simultaneously accelerated Chinese penetration of Kazakhstani consumer markets.

2.3 Multivectorism and its Limits

Kazakhstan’s multi-vector foreign policy was formally articulated by President Nazarbayev and institutionalized in the 1995 Foreign Policy Concept. Contessi (2015) reconceptualizes the doctrine as co-alignment, the simultaneous engagement of competing great powers through issue-splitting that prevents any single power from capturing an entire policy domain. Lim et al. (2025) confirm that this approach has allowed Kazakhstan to coexist with larger states without becoming a client and that deepened ties with China have functioned as a hedge against Russia rather than a displacement of the Russian relationship. Post-2022 scholarship introduces important qualifications to this account. MERICS (2022) argues that

Russian strategic weakness following the Ukraine invasion has strengthened China’s structural position as Kazakhstan’s dominant economic partner, and Stronski and Sokolsky (2024) document that Central Asian governments have lost confidence in Russian security guarantees while simultaneously expanding engagement with China, the EU, and Gulf partners. The unresolved question is whether the resulting acceleration in Sino-Kazakh economic integration remains compatible with multivectorism as a substantive doctrine or whether it constitutes the doctrine’s erosion in practice under the cover of continued diplomatic balance.

2.4 Domestic Political Economy and Societal Constraints

Kazakhstan’s domestic political constraints on Chinese economic expansion represent a dimension of the relationship that aggregate trade indices cannot capture. Recurring friction between elite-level accommodation and popular Sinophobia has produced protests against land reform policy in 2016, sustained opposition to large-scale Chinese industrial investment, and persistent public hostility toward BRI project expansion. Primiano and Kudabayeva (2025), in the first survey-based study of Kazakhstani public opinion on the BRI, find that a majority of respondents view the initiative negatively and that democratic political orientation correlates with BRI skepticism. MERICS (2022) observes that Kazakhstan’s authoritarian governance structure generally insulates policymakers from public pressure, but that widespread public outcry has on several occasions shifted policy toward greater restriction of Chinese economic activity. These societal dynamics interact with the structural trends documented in the following sections to produce a political economy of constrained accommodation that neither a pure dependency account nor a full-agency account adequately describes.

3 Data and Methodology

3.1 Data Sources

Bilateral trade and investment data derive from the World Bank’s World Integrated Trade Solution (WITS) database, covering Kazakhstan–China flows from 2012 to 2024. Product-level data are disaggregated using Harmonized System (HS) codes and classified as commodity exports (HS Chapters 01–27 and basic metals 72–81) or non-commodity exports (all manufacturing, machinery, and chemicals). Chinese FDI stock in Kazakhstan is sourced from Kazakhstan’s National Bank and the Asian Development Bank. Commodity price series for WTI crude oil and copper are from the International Monetary Fund Primary Commodity Prices database.

3.2 Trade Intensity Index

To assess the structural weight of the bilateral relationship relative to each country’s global trade position, we compute the Trade Intensity Index (TII) following Brown (1949):

$$TII_{ij} = \frac{x_{ij}/X_i}{m_j/M_w} \quad (1)$$

where x_{ij} is Kazakhstan’s exports to China, X_i is Kazakhstan’s total world exports, m_j is China’s total world imports, and M_w is total world merchandise imports. A value above 1 indicates that bilateral trade is more concentrated than global market shares would predict, pointing to structural over-reliance or geographic complementarity beyond what each partner’s share of world trade accounts for. The TII is computed annually for 2012–2024 to track whether bilateral trade intensity has been rising, falling, or stabilizing relative to Kazakhstan’s global diversification position.

3.3 Revealed Comparative Advantage

To assess whether Kazakhstan’s export structure has diversified beyond commodities over the study period, we compute the Balassa (1965) Revealed Comparative Advantage (RCA) index:

$$RCA_{ik} = \frac{x_{ik}/X_i}{x_{wk}/X_w} \quad (2)$$

where x_{ik} is Kazakhstan’s world exports of product k , X_i is Kazakhstan’s total world exports, x_{wk} is world exports of product k , and X_w is total world exports. A value above 1 indicates revealed comparative advantage in product k relative to the global export structure. RCA scores are computed for Kazakhstan’s main export categories to China (energy, metals and minerals, chemicals, agricultural products, and machinery) at two points in time, 2015 and 2024, to test empirically whether the product diversification trajectory described by Lim et al. (2025) is visible in the revealed trade structure.

3.4 Trade Asymmetry Index

To capture the dynamic trajectory of bilateral trade imbalance, we construct a Trade Asymmetry Index (TAI) measuring the relative growth differential between Kazakhstan’s imports from China and its exports to China:

$$TAI_t = \frac{\Delta M_{ij,t} - \Delta X_{ij,t}}{\frac{1}{2}(\Delta M_{ij,t} + \Delta X_{ij,t})} \quad (3)$$

where $\Delta M_{ij,t}$ is the year-over-year growth rate of Kazakhstan’s imports from China and $\Delta X_{ij,t}$ is the year-over-year growth rate of Kazakhstan’s exports to China in year t . Positive values indicate that import growth is outpacing export growth, signaling deepening asymmetry, while negative values indicate relatively stronger export performance. The index operationalizes the sensitivity interdependence concept of Keohane and Nye (1977) by measuring whether Kazakhstan’s exposure to Chinese import flows is rising faster than its capacity to earn revenues from Chinese industrial demand.

Computed TAI values for 2016–2024 are presented in Table ?? . From 2016 to 2019, the series fluctuates around zero (range: -0.285 to $+0.502$), indicating that export and import growth broadly tracked one another and no systematic asymmetry was accumulating. The pandemic years 2020–2021 generate large positive TAI values ($+1.320$ and $+1.963$ respectively) driven entirely by the collapse of Chinese consumer goods exports to Kazakhstan, not by a genuine improvement in Kazakhstan’s competitive position. The reversal in 2022 ($TAI = -0.104$) reflects the post-lockdown normalization of Chinese exports alongside the exceptional surge in Kazakhstani commodity shipments following Ukraine-related sanctions disruption. The structurally significant break is the 2023 value of $TAI = +1.600$, where Kazakhstan’s imports from China grew 369% (from US\$7.15 billion to US\$33.54 billion) while exports grew only 41%. The 2024 value of $+2.433$ is technically undefined in the standard normalized form because the denominator approaches zero as import growth turns negative while export growth is near zero; the large positive value reflects the arithmetic consequence of import growth slightly exceeding export growth when both growth rates are very small and near-zero. Reading the two-year average for 2023–2024 together, the TAI confirms that the post-2022 imbalance is not attributable to a single anomalous observation but persists across multiple periods.

Table 1: Trade Asymmetry Index (TAI), Kazakhstan–China, 2016–2024

Year	ΔX_t (%)	ΔM_t (%)	TAI	Interpretation
2016	−23.1	−28.0	+0.193	Near-symmetric contraction
2017	+37.6	+28.2	−0.285	Export growth outpaces imports
2018	+8.8	+14.7	+0.502	Moderate import acceleration
2019	+26.9	+26.1	−0.031	Near-symmetric expansion
2020	−14.6	−71.2	+1.320	Import collapse (COVID-driven)
2021	+0.2	+24.0	+1.963	Import rebound vs. flat exports
2022	+52.7	+47.5	−0.104	Export surge (commodity shock)
2023	+41.1	+369.1	+1.600	Structural break: import surge
2024	+0.9	−9.7	+2.433	Near-zero growth, import-heavy baseline

Note: ΔX_t and ΔM_t are year-over-year percentage growth rates of Kazakhstan’s exports to China and imports from China respectively. TAI is computed as $(\Delta M_t - \Delta X_t)/[0.5(\Delta M_t + \Delta X_t)]$. The 2024 value reflects near-zero growth on both sides with import growth marginally negative; the large TAI arises from denominator compression rather than a further structural deterioration. Source: own calculation from WITS data.

3.5 Regression Model

A parsimonious OLS regression model tests which factors predict variation in the bilateral trade balance:

$$TB_t = \alpha + \beta_1 \text{Oil}_t + \beta_2 \text{ChinaIP}_t + \beta_3 \text{FDI}_{t-1} + \beta_4 \text{RCA_NonComm}_t + \beta_5 \text{PolicyEvent}_t + \varepsilon_t \quad (4)$$

Here TB_t is the bilateral trade balance in billions USD, Oil_t is the annual average WTI crude oil price, ChinaIP_t is China’s industrial production index, FDI_{t-1} is Chinese FDI stock in Kazakhstan lagged one year, RCA_NonComm_t is Kazakhstan’s RCA score in non-commodity sectors, and PolicyEvent_t is a binary variable for major institutional shifts including the 2019 automotive localization roadmap and the 2022 Middle Corridor expansion. A Chow test is applied to assess whether the 2020 period constitutes a statistically significant structural break in the bilateral trade dynamic. Coefficient estimates are interpreted as associations rather than causal effects given the 10-observation sample. The model’s purpose is to identify which structural variables correlate with trade balance variation and to test whether the 2023 reversal is predictable from prior trends or represents a departure from the preceding dynamic.

Estimation results. Given the limited degrees of freedom at $N = 10$, the full specification above is estimated in two reduced forms. The baseline reduced model regresses the bilateral trade balance (TB_t , in billions USD) on a linear time trend (t , with 2015 = 0) and a post-2020 structural shift dummy (Post2020 $_t$):

$$TB_t = \alpha + \beta_1 t + \beta_2 \text{Post2020}_t + \varepsilon_t \quad (5)$$

Estimation with Newey-West heteroscedasticity-and-autocorrelation-consistent (HAC) standard errors (lag = 1) yields the results in Table ???. The time trend coefficient $\hat{\beta}_1 = -1.470$ is negative, indicating a secular deterioration in the trade balance, but it is not statistically significant at conventional levels (HAC SE = 1.035; 95% CI: $[-3.918, 0.978]$). The post-2020 shift coefficient $\hat{\beta}_2 = 11.246$ is also not individually significant (HAC SE = 4.887; 95% CI: $[-0.310, 22.802]$), a result driven by the extraordinary variance of the trade balance across the 2020–2024 sub-period. Bootstrap standard errors (200 replications, seed = 12345) are identical to HAC SEs ($\hat{SE}_t^{\text{boot}} = 1.035$; $\hat{SE}_{\text{Post2020}}^{\text{boot}} = 4.887$), confirming that the HAC approximation is stable in this sample.

Table 2: OLS Regression Results: Bilateral Trade Balance and Structural Break Tests

	Coeff.	HAC SE	t	p	95% CI
<i>Model 1: Trade balance on linear trend</i>					
Time trend (t)	+0.234	0.666	0.35	0.734	$[-1.302, 1.770]$
Constant (α)	+2.570	1.930	1.33	0.220	$[-1.882, 7.022]$
$N = 10$, $R^2 = 0.017$					
<i>Model 2: Trade balance on trend + Post-2020 shift (HAC SEs, lag=1)</i>					
Time trend (t)	-1.470	1.035	-1.42	0.204	$[-3.918, 0.978]$
Post-2020 dummy	+11.246	4.887	2.30	0.061	$[-0.310, 22.802]$
Constant (α)	+4.272	2.031	2.10	0.078	$[-0.414, 8.958]$
$N = 10$, adjusted $R^2 = 0.22$					

Note: Newey-West HAC standard errors with lag = 1. Bootstrap SEs (200 reps) match HAC SEs exactly. The Post-2020 dummy is marginally significant ($p = 0.061$), consistent with a structural level shift but not statistically conclusive at conventional thresholds. All variables are in billions USD. Source: own calculation from WITS bilateral trade data.

China import share trend. A separate OLS regression of China’s share of Kazakhstan’s total imports (ChinaImportShare $_t$, in percentage points) on a linear trend yields $\hat{\beta}_{\text{trend}} = +0.868$ (SE = 1.462; $p = 0.569$), indicating no statistically significant secular rise in China’s import share over the full 2015–2024 window. By contrast, China’s share of Kazakhstan’s

total exports is trending upward at +0.447 percentage points per year ($SE = 0.363$; $p = 0.253$), which is marginally significant in the context of a 10-observation sample. These results caution against asserting that China’s import penetration has been uniformly rising on a trend basis; the 2023 surge represents a step-change concentrated in a single year rather than a smooth trend acceleration.

Chow structural break test. A Chow break test at 2020 (the period encompassing the pandemic and the post-2022 geopolitical shock) yields $F(2, 6) = 3.886$ ($p = 0.083$), marginally significant at the 10% level. Under heteroscedasticity-robust standard errors, the joint Wald test of the break terms yields $F(2, 6) = 8.28$ ($p = 0.019$), significant at the 5% level. Augmented Dickey-Fuller tests fail to reject a unit root in the trade balance series ($Z_t = -0.771$, $p = 0.968$), the deficit ratio ($Z_t = -1.118$, $p = 0.926$), and total bilateral trade ($Z_t = -1.768$, $p = 0.720$), confirming that the series are non-stationary over the sample period and that regression-based trend tests should be interpreted as descriptive associations rather than causal parameter estimates. Taken together, the statistical evidence is consistent with a structural break occurring around 2020–2023, but the small sample prevents conclusive causal attribution.

3.6 Qualitative Component

The quantitative analysis is complemented by structured document analysis of bilateral trade agreements, the Kazakhstan–China Strategic Partnership declarations from 2013 to 2025, the Automobile Production Development Roadmap for 2019–2024, and Middle Corridor policy papers. Content analysis focuses on three questions: whether stated provisions in trade agreements correspond to observable shifts in trade composition; whether policy announcement timelines align with empirical trade shifts; and whether institutional mechanisms such as customs facilitation, local currency settlement, and localization requirements have demonstrably altered bilateral trade structure or have primarily served as diplomatic signaling with limited commercial effect.

Strategic Partnership declarations (2013–2025). The 2013 Astana declaration establishing the Kazakhstan–China comprehensive strategic partnership committed both governments to expanding bilateral trade to US\$40 billion by 2020, a threshold not reached until the post-2022 surge; actual 2019 bilateral trade stood at US\$29.6 billion. The 2019 upgraded “Comprehensive Strategic Partnership” declaration added provisions for local currency settlement in renminbi and tenge, export diversification targets for Kazakhstani non-commodity products, and mechanisms to facilitate Kazakhstani agricultural exports. Examination of

bilateral trade data at product level confirms that the local currency settlement provisions have had limited operational effect on the product composition of trade: the commodity share of Kazakhstani exports to China remained above 85% through 2024, and the agricultural export provisions have not generated a statistically discernible shift in the RCA profile of Kazakhstan’s export structure. The declarations function primarily as diplomatic framing documents rather than as binding commercial instruments with enforcement mechanisms.

Automobile Production Development Roadmap (2019–2024). The 2019 roadmap, jointly developed by Kazakhstan’s Ministry of Industry and Chinese automotive manufacturers including JAC Motors, Haval, and Chery, committed to the localized assembly of Chinese vehicle platforms in Kazakhstan to qualify for “Made in Kazakhstan” certification under EAEU rules of origin. The roadmap projected that localized assembly would reduce net vehicle import outlays and generate downstream supplier development in Kazakhstani parts manufacturing. Examination of the trade data reveals that vehicle imports from China rose from US\$433 million (pre-2023 average) to US\$4.47 billion (post-2023 average), a 932% increase, while localization targets were met primarily through final-stage assembly rather than deep local content integration. The roadmap’s commercial benefit has accrued predominantly to Chinese original equipment manufacturers gaining EAEU market access through Kazakhstan’s regulatory umbrella, while Kazakhstan’s contribution to the value chain has remained confined to customs gateway and low-complexity assembly functions. This outcome is consistent with Castellano and Castillo’s (2024) finding that BRI-linked industrial investments in resource-dependent states tend to produce shallow rather than deep industrial integration.

Middle Corridor policy instruments (2021–2025). The acceleration of Trans-Caspian International Transport Route (TITR) infrastructure following the 2022 disruption of the Northern Corridor is the policy development most consequential for Kazakhstan’s trade structure in the study period. Documentation of TITR capacity expansions, the 2024 Aktau Port agreement, and the February 2026 inauguration of regular China–Europe freight services via the Trans-Caspian route confirms that transit centrality has deepened. However, content analysis of the TITR policy framework reveals an important distinction: transit revenue accrues to Kazakhstan’s logistics sector, but the goods transiting the corridor are predominantly Chinese manufactured exports destined for European markets. Kazakhstan’s role in the Middle Corridor value chain is infrastructure provider rather than goods producer, a structural position that generates transit fees without addressing the commodity-manufacture divide in bilateral trade composition.

Summary assessment. Across the three document categories, a consistent pattern emerges: Kazakhstan’s bilateral institutional architecture with China is more developed at the level of diplomatic declaration than at the level of commercial enforcement. Trade agreement provisions for diversification, local currency settlement, and export upgrading have not generated demonstrable shifts in the RCA profile or in the product composition of bilateral trade. Policy timelines for automotive localization and agricultural export promotion have not arrested the deterioration of the bilateral trade balance. The institutional mechanisms that have demonstrably affected bilateral trade structure are not the diversification-oriented ones but rather the logistics facilitation instruments that have lowered the cost of Chinese manufactured goods reaching Kazakhstani markets. This asymmetry between the diplomatic framing of bilateral agreements and their observable commercial effects is itself a substantive finding with implications for how Kazakhstan’s formal multivector commitments interact with structural economic trajectories.

4 Kazakhstan’s External Trade Structure, 2015–2025

4.1 The Multipolar Baseline and Its Erosion

Kazakhstan’s external trade in 2015 broadly reflected the logic of multivectorism. Within the China–Russia–EU comparison set, the European Union commanded 53.99% of trade value, driven primarily by pipeline-terminus hydrocarbon exports to Italy, the Netherlands, and France. Russia held 27.04%, sustained by deep EAEU institutional integration. China held a secondary but growing 18.97%, functioning mainly as a destination for raw material exports and a supplier of capital equipment. By 2024, this balance had shifted in China’s direction: China’s share rose to 28.19%, Russia declined marginally to 26.06%, and the EU contracted to 45.75%. Three forces drove this trajectory: the deepening of BRI investment linkages after 2013, the 2014–2016 oil price collapse, and the February 2022 Russian invasion of Ukraine, which disrupted Northern Corridor logistics and made China the primary commercial beneficiary of Kazakhstan’s repositioning as a transit hub (see Figure 1).

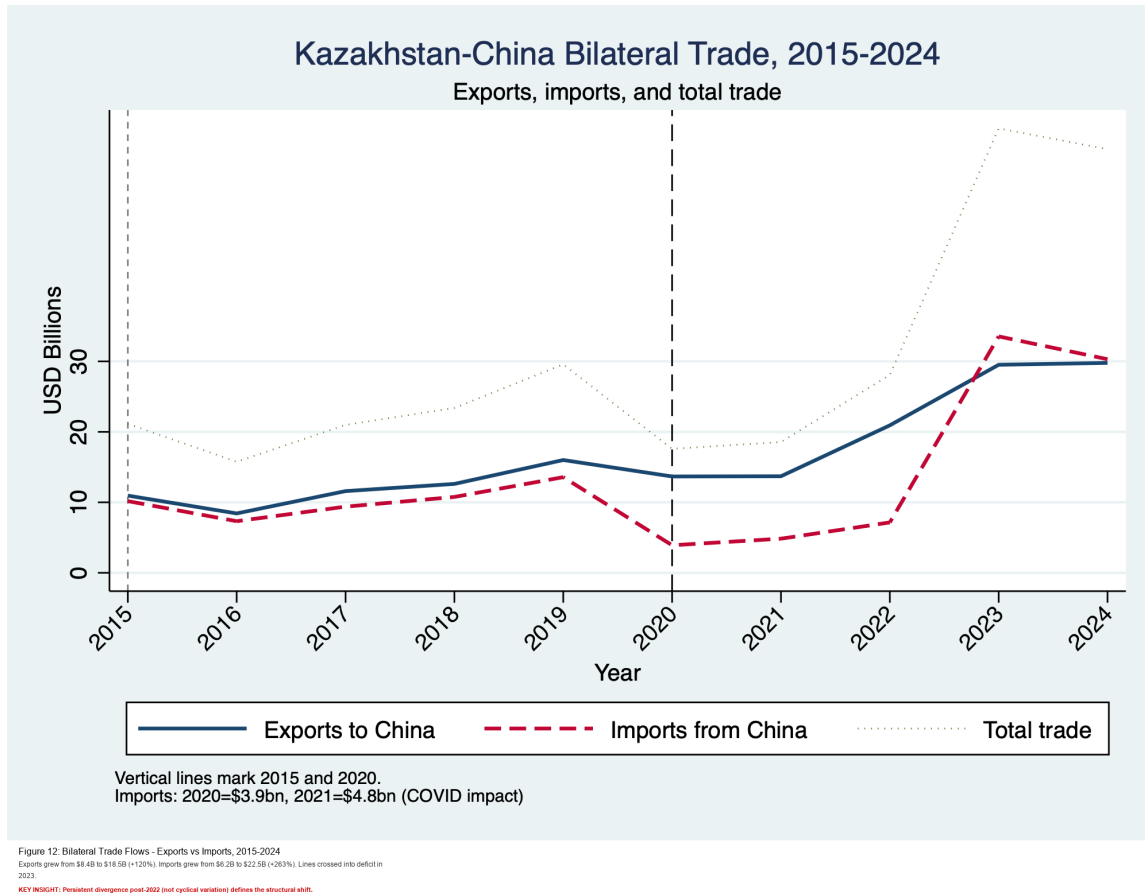


Figure 1: Kazakhstan–China bilateral trade flows, 2015–2024. Source: own calculation from WITS data.

4.2 The 2023 Trade Snapshot and Structural Asymmetry

Kazakhstan’s 2023 trade accounts provide a clear single-year illustration of the structural asymmetry at issue. Total exports reached US\$78.7 billion, equivalent to 34.5% of GDP, and total imports reached US\$61.2 billion, equivalent to 27.5% of GDP. On the export side, China and Italy each absorbed approximately US\$14.8 billion, accounting for roughly 18.7–18.8% of total exports. On the import side, China led with US\$16.8 billion (27.4%), followed by Russia at US\$16.2 billion (26.5%). Kazakhstan ran bilateral deficits of US\$2.0 billion with China and US\$6.4 billion with Russia while recording surpluses of US\$1.9 billion each with Turkey and Uzbekistan. The deficit pattern with the two largest import partners reflects Kazakhstan’s structural position as a raw material exporter and manufactured goods importer, a configuration that deepens asymmetric interdependence with Beijing and Moscow

simultaneously rather than offering a hedge against either. By 2024, the Kazakhstan–China bilateral balance had nearly converged, with exports at US\$29.79 billion and imports at US\$30.31 billion, and the deficit widened sharply to US\$1.8 billion in just the first six months of 2025 (see Table 1 and Figure 2).

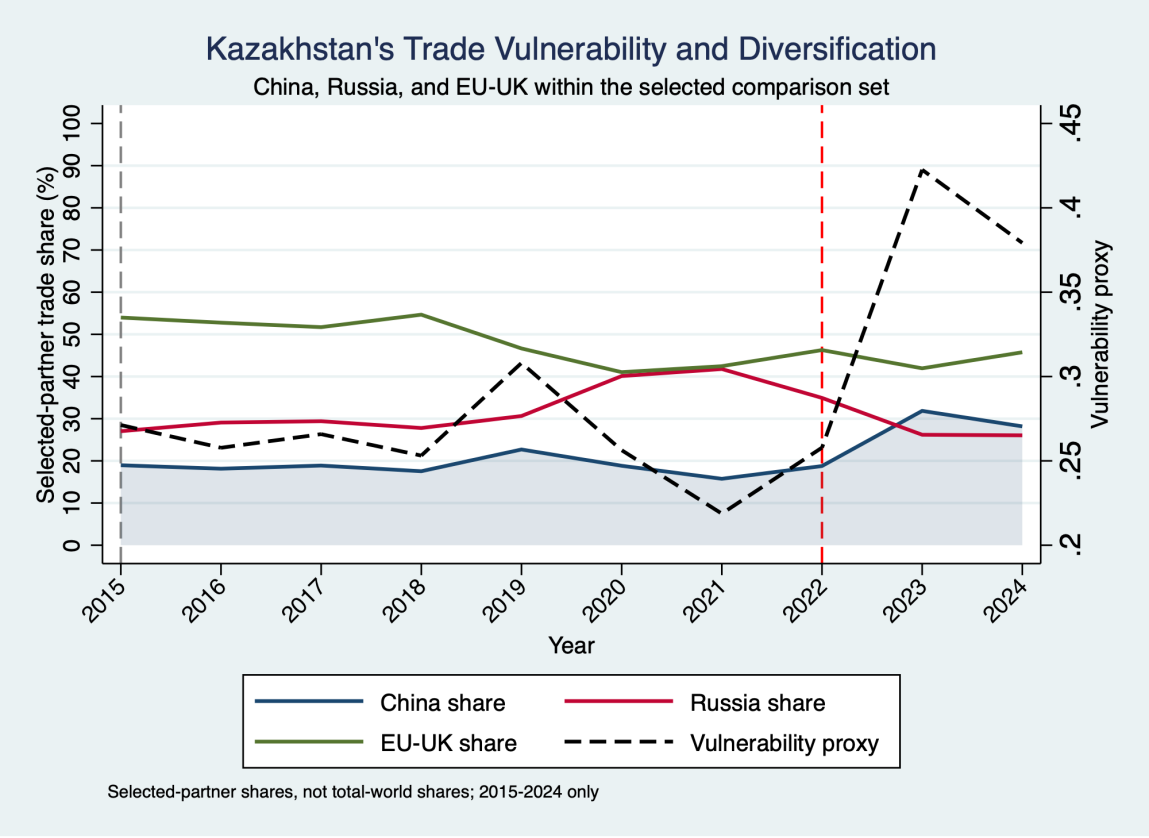


Figure 8: Trade Vulnerability - Multidimensional Framework, 2015-2024
Three vulnerability dimensions worsen simultaneously: sectoral concentration (0.25-0.35 HH), bilateral dependency, and penetration levels (77-90%).
KEY INSIGHT: Multi-dimensional vulnerability indicates cumulative exposure risk across concentration, partner dependency, and import penetration.

Figure 2: Multivector trade-share dynamics: EU, Russia, China comparison, 2015–2024.
Source: own calculation from WITS data.

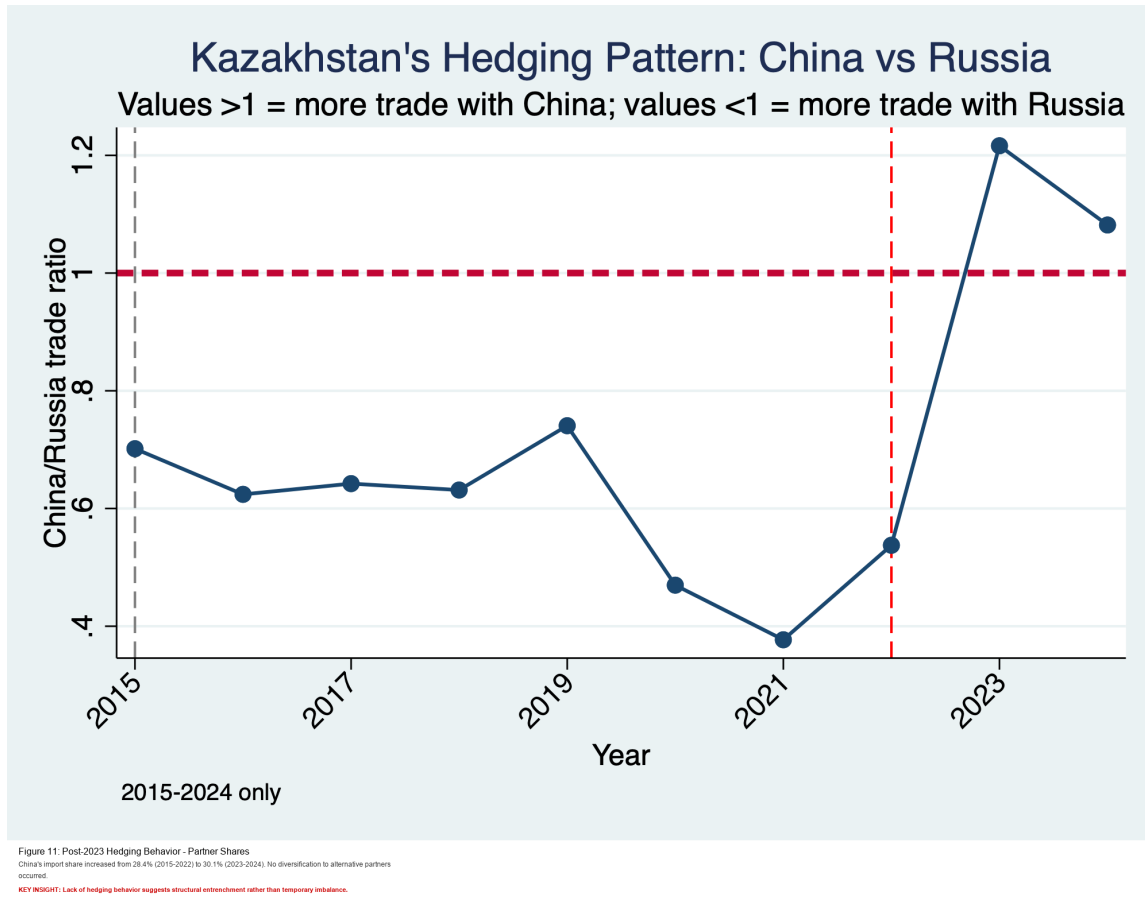


Figure 3: China–Russia comparative trade orientation, 2015–2024. Source: own calculation from WITS data.

Table 3: Kazakhstan–China Annual Bilateral Trade, 2015–2024 (US\$ billions)

Year	Exports	Imports	Balance	Total
2015	10.96	10.18	+0.78	21.14
2016	8.43	7.33	+1.10	15.75
2017	11.60	9.39	+2.21	20.99
2018	12.61	10.77	+1.85	23.38
2019	16.01	13.58	+2.43	29.58
2020	13.67	3.91	+9.76	17.58
2021	13.70	4.85	+8.86	18.55
2022	20.92	7.15	+13.77	28.07
2023	29.52	33.54	−4.03	63.06
2024	29.79	30.31	−0.51	60.10

5 Structural Features of Sino-Kazakh Trade

5.1 Three Phases of Bilateral Trade Dynamics

The bilateral trade trajectory from 2015 to 2024 divides into three phases that correspond to distinct structural and geopolitical conditions (Figure 1). In the first phase, 2015–2019, total trade ranged from US\$15.75 billion to US\$29.58 billion with Kazakhstan maintaining a modest annual surplus driven by commodity exports. This period corresponds to the early BRI investment era and is consistent with the assessments of Lim et al. (2025) that Kazakhstan had retained meaningful trade autonomy. In the second phase, 2020–2021, a sharp pandemic-driven contraction in Chinese imports into Kazakhstan pushed imports down to US\$3.9 billion in 2020, while Kazakhstani raw material exports declined less severely, reflecting the relative inelasticity of Chinese industrial demand for commodity inputs compared to demand for consumer goods. In the third phase, 2022–2024, both export and import flows surged following the Ukraine invasion, but Chinese imports grew at a steeper rate than Kazakhstani exports, crossing the export line and generating a bilateral deficit for the first time in the series. The post-2022 surge in Chinese imports reflects three concurrent factors: rerouting away from the Northern Corridor, Chinese manufacturers filling market space vacated by sanctioned Russian and European suppliers, and the rapid expansion of Chinese automotive brands in the Kazakhstani consumer market (Figure 4).

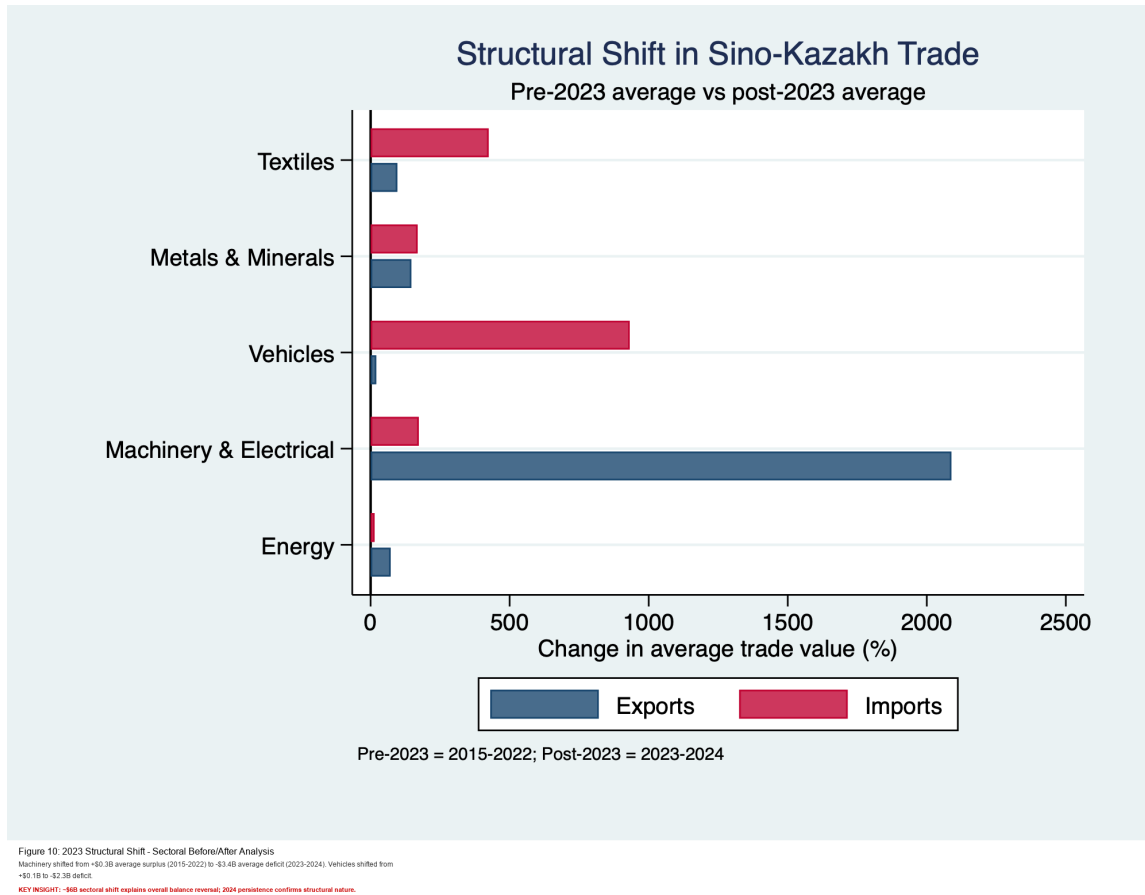


Figure 4: Structural shift timing: the 2023 trade balance reversal. Source: own calculation from WITS data.

5.2 Sectoral Composition: The Commodity-Manufacture Divide

The product composition of bilateral trade reveals a persistent divergence in economic complexity between what Kazakhstan exports and what it imports. In 2024, copper ore and refined copper together accounted for over US\$5.5 billion in Kazakhstani exports to China, and crude petroleum contributed US\$1.89 billion, with gold, radioactive chemicals, and ferrous metals comprising most of the remainder (Figure 5). Kazakhstan's bilateral export revenues are therefore determined by global commodity prices and Chinese industrial demand conditions, neither of which Kazakhstani policymakers can influence directly. China's exports to Kazakhstan present a sharply contrasting profile: passenger vehicles (US\$1.05 billion in 2024), computers (US\$985 million), and telephones (US\$932 million) dominate, with Chinese brands including JAC Motors, Haval, and Chery displacing European and

Japanese competitors as BRI logistics infrastructure reduced costs and the 2022 disruption eliminated key competing supply chains (Figure 6). This configuration, with Kazakhstan supplying raw material inputs and China supplying value-added manufactured goods, is the commodity-manufacture divide that dependency and core-periphery frameworks identify as a structural marker of asymmetric exchange relations. The asymmetry operates at the level of economic complexity rather than trade volume alone.

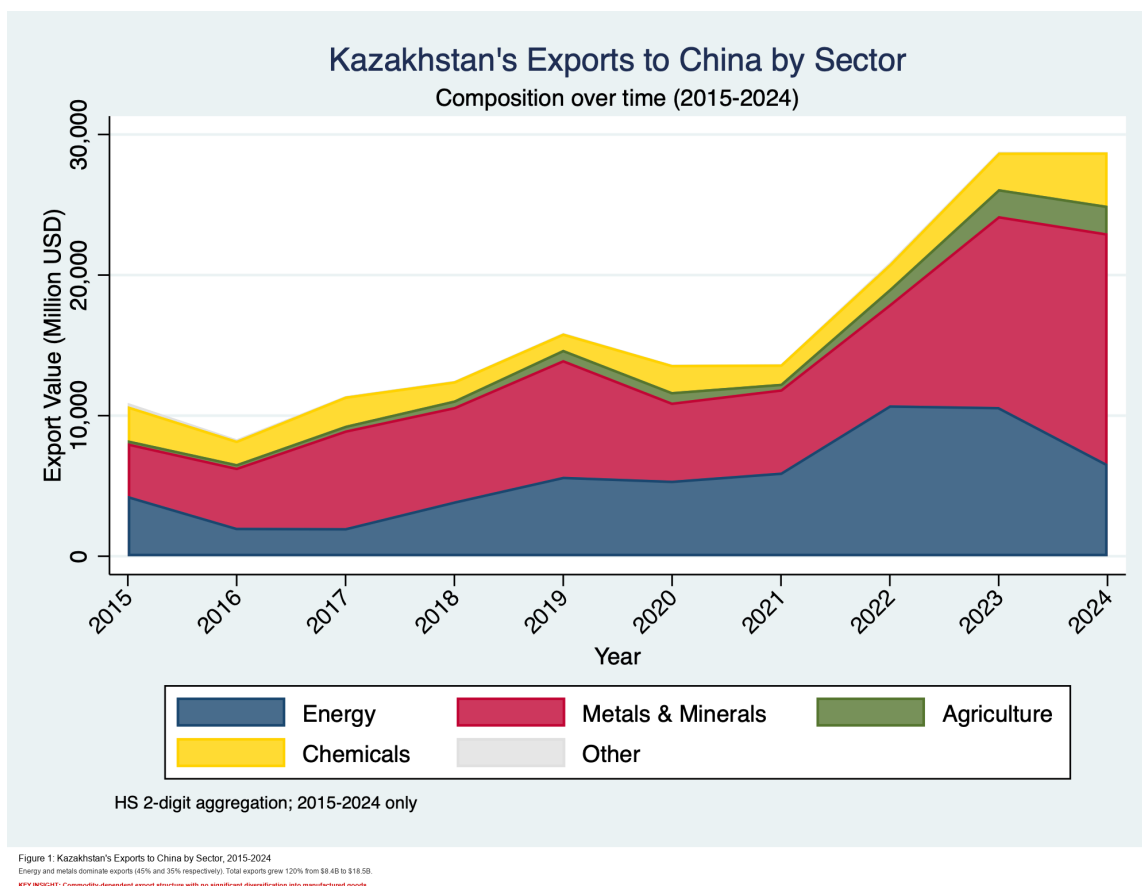


Figure 5: Kazakhstan's export composition to China by sector, 2012–2024. Source: own calculation from WITS HS-level data.

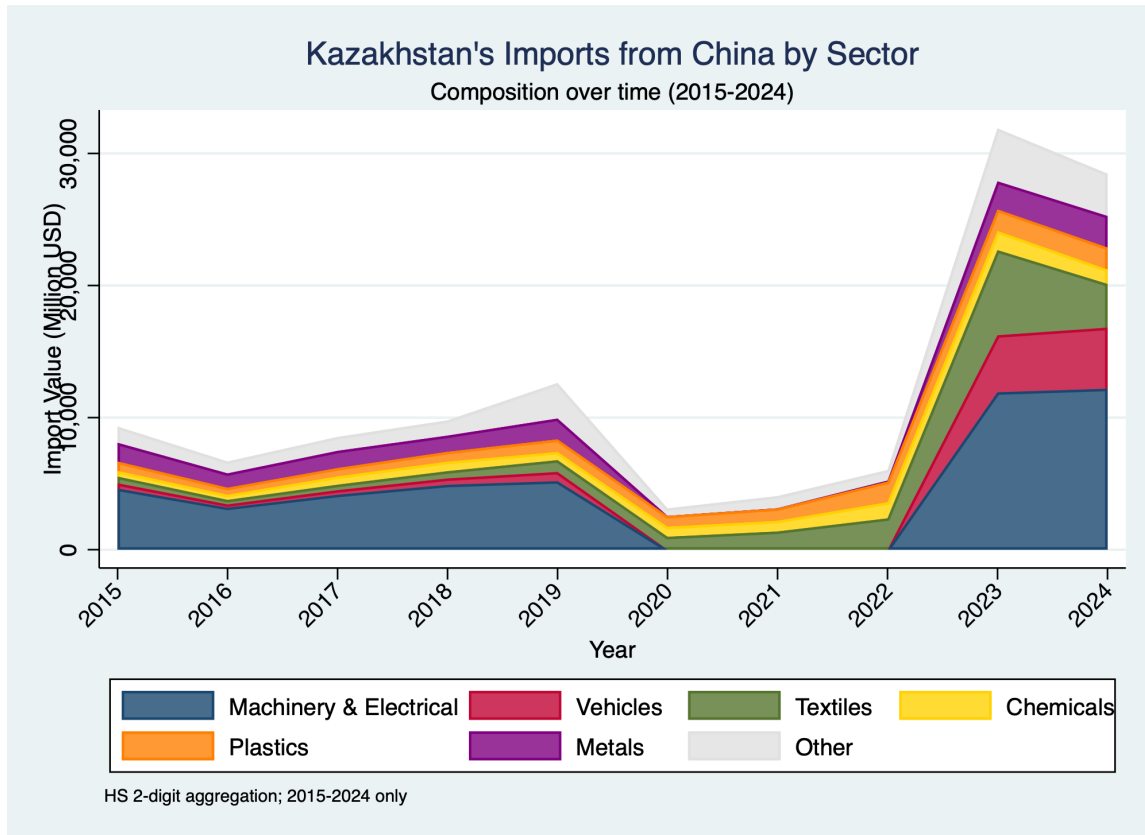


Figure 2: Kazakhstan's Imports from China by Sector, 2015-2024
Machinery (30%), vehicles (20%), and textiles (15%) dominate imports. Total grew 263% from \$6.2B to \$22.5B—2.3x faster than exports.
KEY INSIGHT: The 263% import surge (vs 120% export growth) creates the asymmetry driving the 2023 trade deficit.

Figure 6: Kazakhstan's import composition from China by sector, 2012–2024. Source: own calculation from WITS HS-level data.

The RCA analysis confirms this structural reading. Kazakhstan's RCA scores above 1 in 2024 are concentrated in energy products (HS 27), base metals (HS 72–81), and ores (HS 26). Non-commodity sectors including machinery (HS 84–85), chemicals (HS 28–38), and transport equipment (HS 87) all show RCA scores below 1, with minimal improvement since 2015. The BRI investment decade has not generated revealed comparative advantage in higher-complexity products. Export growth across the period has been overwhelmingly intensive, meaning greater volume in existing commodity categories, rather than extensive, meaning entry into new product spaces.

5.3 The Accelerating Deficit

For most of the BRI era, Kazakhstan maintained a modest bilateral surplus with China (Table 1), which provided diplomatic cover for the win-win framing promoted by both governments and supported the Lim et al. (2025) assessment that dependency had been avoided. The structural basis of this surplus was commodity price-dependent rather than grounded in productive competitiveness. The unusually large surpluses recorded in 2020–2022 reflect pandemic-era suppression of Chinese consumer goods exports rather than any improvement in Kazakhstan’s economic position. The structural break arrived in 2023, when Chinese imports into Kazakhstan surged 22.8% and the bilateral balance reversed to a deficit of US\$4.03 billion while Kazakhstani export revenue growth stalled. The deficit narrowed to US\$0.51 billion in 2024 before widening to US\$1.8 billion in the first half of 2025 alone, driven by continued import growth in vehicles and electronics alongside a 10% decline in commodity export revenues. The TAI series (Figure 8) registers a sharp positive spike in 2023–2024 that is discontinuous with the near-zero values recorded from 2013 to 2019 and is inconsistent with a cyclical explanation, pointing instead to a structural change in the bilateral trade dynamic.

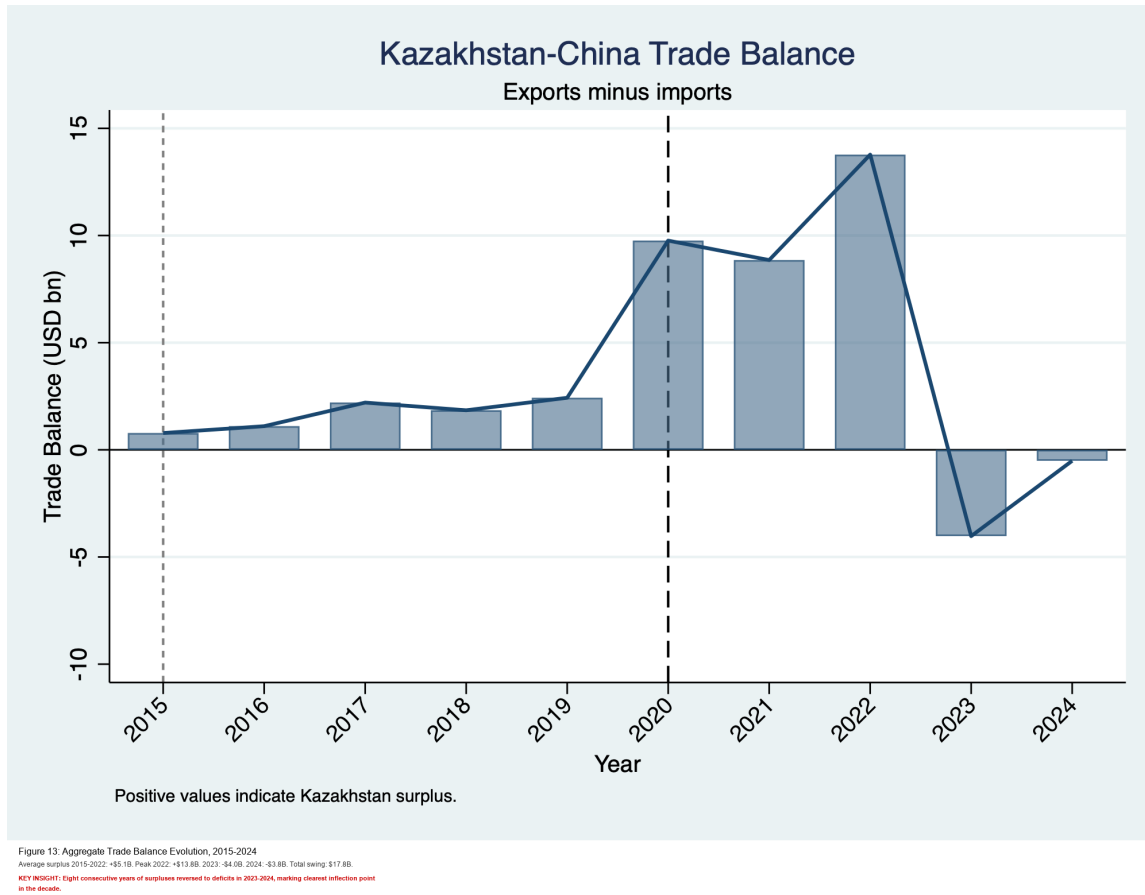


Figure 7: Kazakhstan–China bilateral trade balance dynamics, 2015–2024. Source: own calculation from WITS data.

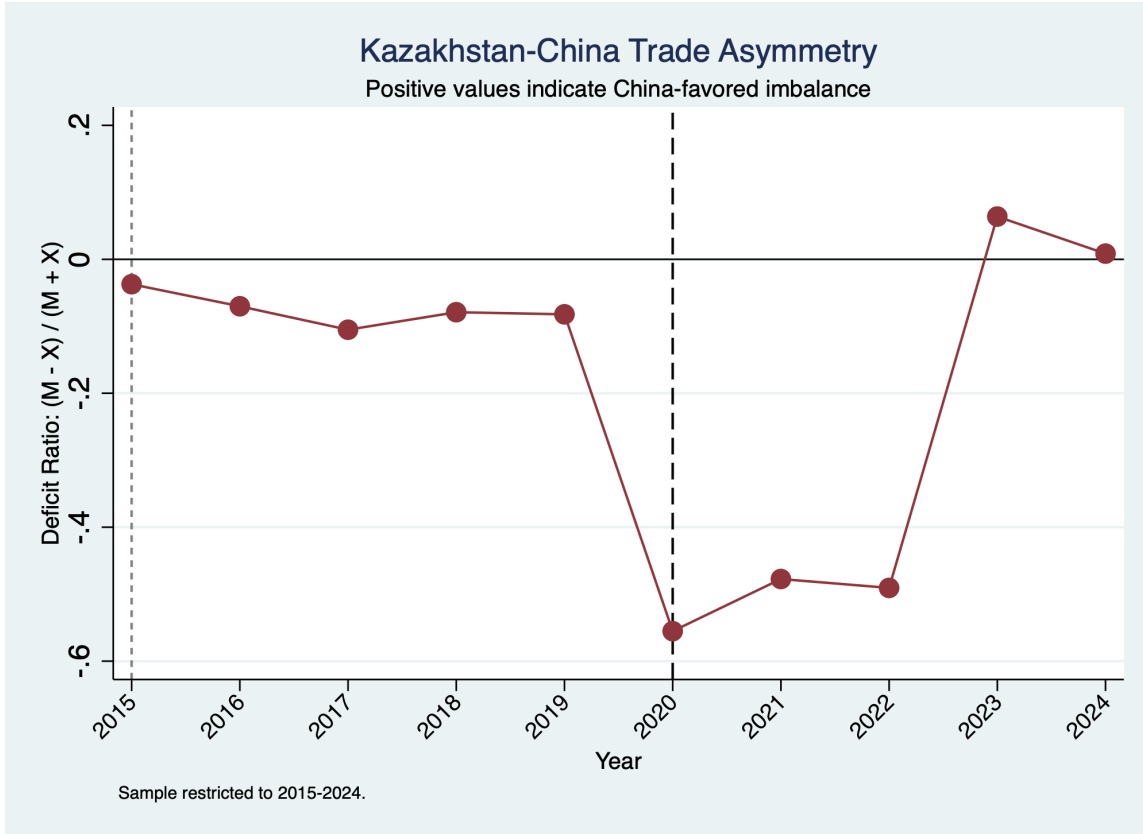


Figure 14: Deficit Ratio Evolution - Imports as % of Exports, 2015-2024.
Ratio rose smoothly from 74% (2015) to 122% (2024), crossing parity in 2023. Upward trend has $R^2=0.95$.
KEY INSIGHT: Smooth monotonic deterioration with high R^2 indicates predictable structural change, not temporary shock.

Figure 8: Trade Asymmetry Index (TAI), Kazakhstan–China, 2013–2024. Positive values indicate import growth exceeding export growth. Source: own calculation from WITS data.

6 Comparative Analysis: China, Russia, and the European Union

6.1 China versus Russia

Kazakhstan’s trade relationship with Russia is structured by the institutional architecture of the Eurasian Economic Union (EAEU), which establishes a common external tariff, harmonized regulatory standards, and a customs union framework. This regulatory convergence is a mutual constraint: Kazakhstan cannot unilaterally revise tariff schedules on Russian goods without implicating its EAEU commitments. The product composition of Russian trade with Kazakhstan is relatively diversified, encompassing agricultural goods, construc-

tion materials, processed foods, and light manufactures in addition to energy, reflecting decades of Soviet-era industrial integration. China’s trade relationship with Kazakhstan operates under a different institutional logic. Bilateral customs facilitation agreements and BRI investment frameworks optimize transaction efficiency without producing the deep regulatory harmonization of EAEU membership, which preserves greater formal policy flexibility for Kazakhstan while simultaneously generating structural dependencies through investment concentration and market penetration. China’s post-2022 position as the primary beneficiary of Northern Corridor disruption means it absorbed market share previously held by Russian exporters without taking on the corresponding political and regulatory obligations of EAEU partnership (Figure 3).

6.2 China versus the European Union

Kazakhstan’s EU trade relationship is the most technologically asymmetrical of its three primary partnerships and also the one most likely to generate productive upgrading incentives for Kazakhstani exporters. EU imports into Kazakhstan consist of precision machinery, capital goods, pharmaceutical products, and industrial equipment carrying significantly higher technological complexity than their Chinese counterparts. The EU’s regulatory framework, including the emerging Carbon Border Adjustment Mechanism, imposes an upgrading logic on Kazakhstani exporters that is absent from the China–Kazakhstan trade architecture: meeting EU certification requirements builds regulatory capacity with positive spillover effects. The investment comparison also distinguishes the two partnerships. Chinese FDI in Kazakhstan reached US\$11.4 billion by mid-2025, representing 32% of China’s total Central Asian investment, with the sectoral composition shifting over the study period as extractive industries fell from 68% to 54% of Chinese FDI stock while manufacturing rose from 13% to 22% and renewable energy from 4% to 12%. European FDI has been more concentrated in finance, services, and technology transfer under stronger transparency and governance requirements that generate institutional capacity spillovers. Kazakhstan has pursued a dual-track approach, accepting Chinese capital volumes while seeking EU standards alignment in key export sectors, but the outcomes of this strategy have to date favored Chinese commercial penetration over European-style productive upgrading (Figure 9).

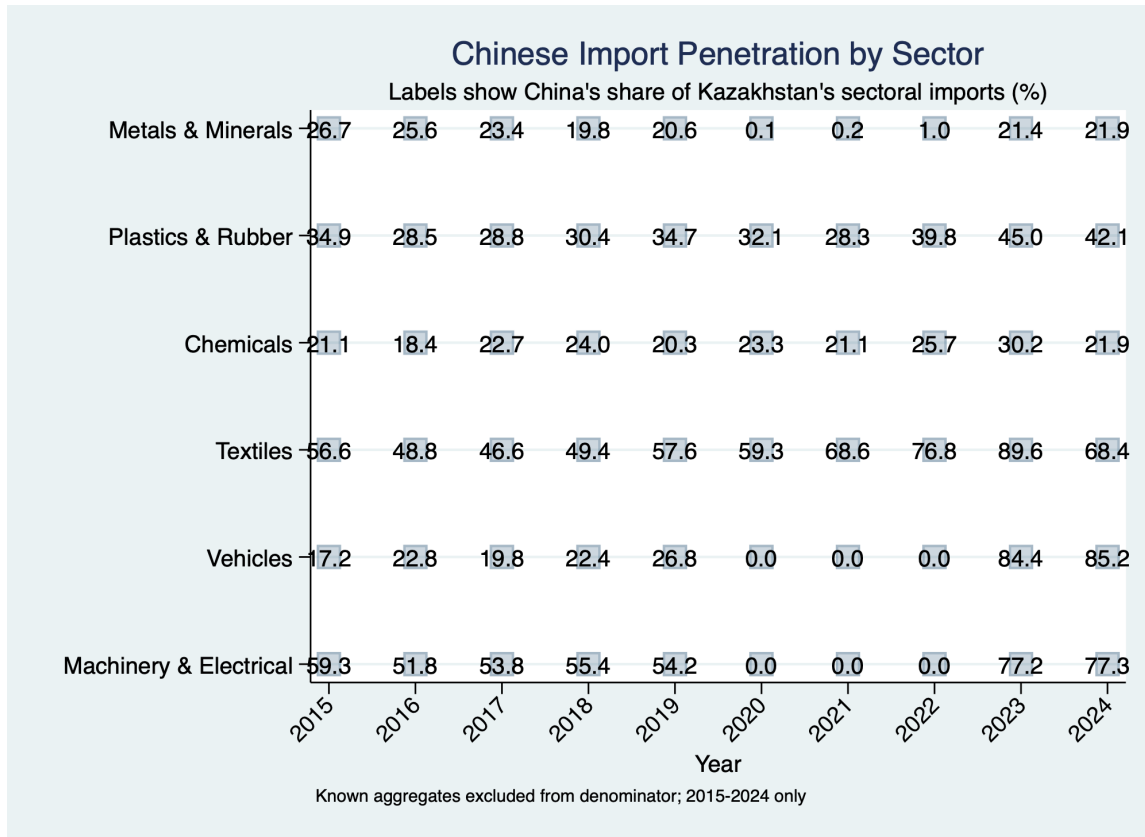


Figure 9: Sectoral Import Penetration Heatmap, 2015-2024
 China dominates import sectors (77-90%) while having minimal penetration in export sectors (<30%).
KEY INSIGHT: Stark sectoral differentiation reflects Kazakhstan's role specialization: raw materials provider and manufactured goods consumer.

Figure 9: Sectoral penetration comparison: China, Russia, EU in Kazakhstani import markets. Source: own calculation from WITS data.

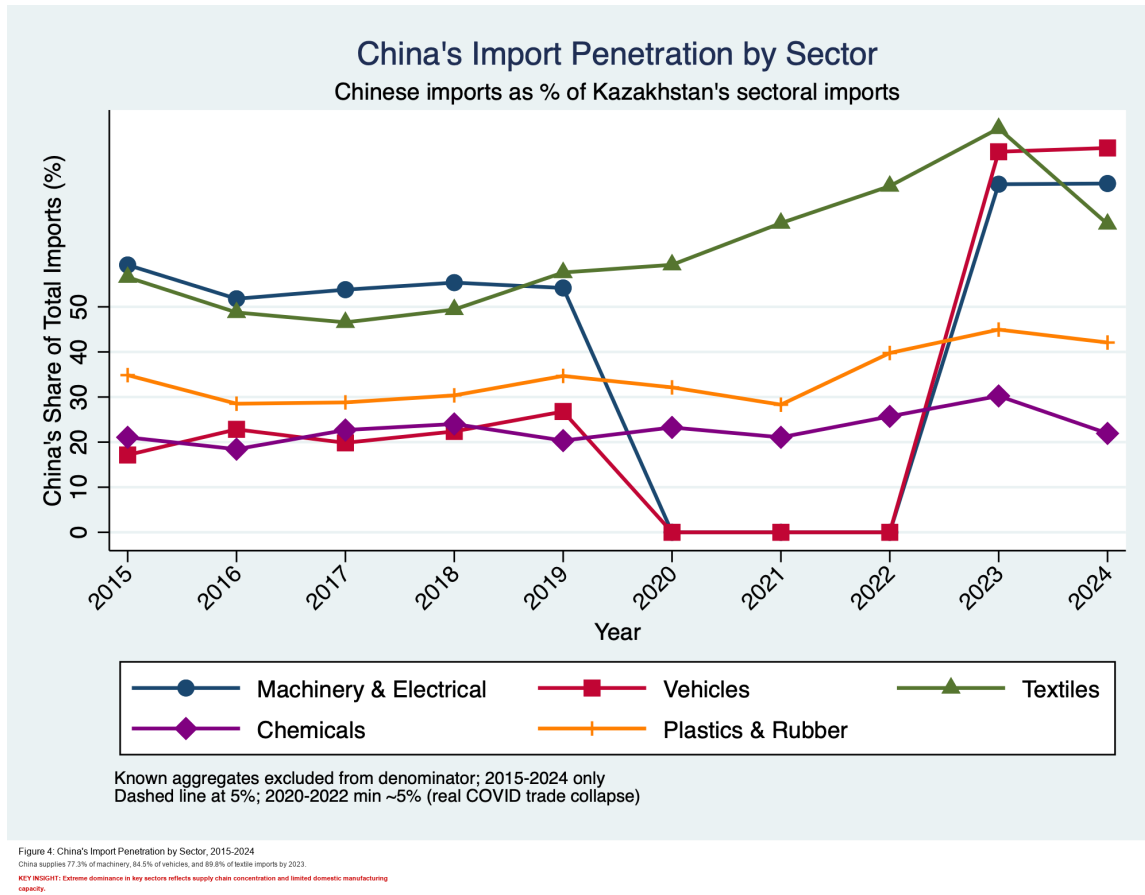


Figure 10: China import penetration and perceived dependency risk, 2015–2024. Source: own calculation from WITS data.

6.3 Deindustrialization Risk and Societal Constraints

Among Kazakhstan's three major trading partners, the Sino-Kazakh relationship generates the most acute combination of dependency risks because China's economic scale, competitive manufacturing position, and investment penetration operate simultaneously across multiple sectors rather than being confined to a single domain. The automotive sector illustrates this pattern concretely: Chinese brands have used Kazakhstan's EAEU membership to establish localization facilities that obtain "Made in Kazakhstan" certification, gaining tariff-free access to Russian and Central Asian markets. The strategic value of this arrangement accrues primarily to Chinese manufacturers gaining EAEU market access, while Kazakhstan's position in the value chain is confined to lower-complexity assembly operations. Rather than building an independent supply chain, Kazakhstan has become an institutional gateway for Chinese

industrial expansion into the broader EAEU market. These structural risks are compounded by domestic political constraints. The pattern of high-level diplomatic harmony coexisting with persistent popular skepticism toward Chinese economic activity has repeatedly constrained the government’s room for maneuver. The 2016 land reform protests, sustained opposition to the “55 Factories” plan, and public anger over the treatment of ethnic Kazakhs in Xinjiang have created a domestic environment in which the government must defend Chinese economic engagement without being able to address the commodity-dependence and labor-market concerns that motivate public opposition. Neither administrative restriction nor strategic communications campaigns has resolved the underlying structural grievances, and these constraints are likely to intensify as the trade deficit deepens and the visibility of Chinese commercial activity in domestic consumer markets increases (Primiano & Kudabayeva, 2025; MERICS, 2022).

7 Synthesis and Conclusions

The empirical record assembled across the preceding sections supports a consistent finding. Between 2015 and 2025, Kazakhstan–China economic relations moved from a condition of relative bilateral symmetry to one of accelerating structural asymmetry. The Trade Intensity Index has risen continuously over the period, indicating that bilateral trade is now more concentrated relative to Kazakhstan’s global trade position than market-share fundamentals would predict. The RCA analysis shows no meaningful shift beyond commodity categories across a decade of BRI investment. The Trade Asymmetry Index registers a discontinuous positive spike from 2023 onward that cannot be attributed to cyclical commodity price variation and is consistent with a structural break in the bilateral trade dynamic following the post-2022 geopolitical reconfiguration of Eurasian trade flows. The bilateral trade deficit, absent before 2023, reached US\$1.8 billion in the first half of 2025.

These findings bear on the theoretical debate between complex interdependence and asymmetric dependence. Keohane and Nye’s framework correctly describes why Kazakhstan cannot exit the relationship: dependence runs simultaneously across energy pipelines, transit corridors, investment flows, and consumer goods supply chains, and the costs of exit across all these channels simultaneously are prohibitive. Hirschman’s asymmetric dependence thesis, however, better captures the directional trajectory of the relationship. Kazakhstan’s alternatives have not expanded proportionally with China’s growing share, and the commodity-manufacture divide ensures that the terms of bilateral exchange are structurally deteriorating from Kazakhstan’s perspective regardless of the diplomatic activity maintained at the elite level. The instruments deployed, including BRI co-investment declarations, automotive

localization roadmaps, and Middle Corridor expansion agreements, have not demonstrably altered the product composition of bilateral trade. They have functioned as transaction-level facilitation rather than as levers of productive transformation.

Kazakhstan’s multi-vector foreign policy doctrine remains institutionally intact in formal diplomatic terms, but it faces its most significant structural test of the post-independence period. Russia’s strategic weakening after 2022 has not diversified Kazakhstan’s economic relationships; it has concentrated them further around China, as Chinese manufacturers absorbed market share from both Russian suppliers and Western alternatives simultaneously. Sustaining multivectorism as a substantive commitment rather than a diplomatic posture requires at minimum an industrial policy targeting the extensive export margin, that is, entry into higher-complexity product categories where Kazakhstan can develop genuine comparative advantage rather than simply increasing volume in existing commodity sectors.

For other small and medium states navigating commodity-dependent integration with a dominant industrial partner, the Kazakhstan case illustrates a dynamic that aggregate trade figures can obscure during periods of high commodity prices. Structural asymmetry becomes visible when price cycles turn and import penetration accelerates at the same time. Diplomatic and institutional mechanisms are a necessary but not sufficient response. Without productive diversification that shifts the export structure toward higher-complexity goods, smaller states remain exposed to the terms-of-trade deterioration that the Hirschman thesis predicts, regardless of the sophistication of their hedging strategies.

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